Demand forecasting is just predicting how much of something customers will want in the future.

Think of it like this:

- You run a small ice-cream shop. Last month you sold 200 cones. Will you sell 180, 220, or 300 next month? Forecasting gives you a smart guess so you don't run out of cones (customers angry) or buy too many (money wasted on melting ice-cream).

Why do companies care?

- Grocery stores forecast milk so shelves are full but nothing expires.
- Phone makers forecast how many new phones people will buy so they don't over-produce and lose money.
- Hospitals forecast medicine demand so patients don't wait.

We use past sales data (called historical demand) to make the guess. Today we learned two easy methods: Moving Average and Exponential Smoothing. We also learn about two important patterns in the data: trend and seasonality.

A. Moving Average (MA)

- **Simple idea**: Take the average of the last few months' actual sales and use that as your forecast for next month. It "smooths" out random ups and downs.
- **3-month Moving Average** (what we'll use):
- Formula (in math terms):

  $$F_t = \frac{D_{t-1}+D_{t-2}+D_{t-3}}{3}$$

  where F_t = forecast for this month, D = actual demand.
- **Real-life example**: Imagine an umbrella shop. Rainy months sell more. But if you just average the last 3 months (some rainy, some sunny), it gives a calm "average" number so you don't over-buy umbrellas on a random sunny week.
- **Good for**: Stable demand with random noise (no big changes).
- **Weakness**: It always lags behind big changes (slow to react).

B. Exponential Smoothing (ES)

- **Simple idea**: Same as moving average, but it gives **more weight to recent months** instead of treating all months equally.
- You control how much weight with a number called **α (alpha)** between 0 and 1.
  - $\alpha = 0.2$ → very smooth, trusts old data a lot (slow reaction).
  - $\alpha = 0.5$ → reacts faster to recent changes.

- Formula (this is the key one):
  $$F_{t+1} = \alpha \times D_t + (1 - \alpha) \times F_t$$
  (New forecast = α × latest actual + (1-α) × previous forecast)
- **Real-life example**: Coffee shop forecasting daily coffee sales. Yesterday's sales matter more than sales from 3 weeks ago (maybe the weather changed or a new competitor opened). With α=0.5 it reacts quicker than α=0.2.
- **Good for**: When recent data is more important.

C. Trend vs Seasonality (two big patterns you must know)

- **Trend**: A long-term, steady direction the demand is going (up or down over many months/years). Example: Electric cars – demand has been going **up** every year for the last 5 years because more people want eco-friendly cars. Not because of any season – it's a permanent shift.
- **Seasonality**: Regular, repeating pattern that happens at the **same time every year**. Example: Ice-cream sales – they always go **up** every summer (June–August) and down in winter. It repeats every single year like clockwork. Another example: Toy sales explode in December every year because of Christmas.
- **Why they matter**: Simple Moving Average and Exponential Smoothing work okay on flat data, but they struggle when strong trend or seasonality exists (they lag or miss the peaks). That's why later days you'll learn better methods (like Holt-Winters) that separate trend + seasonality.

Related quick concept (just for understanding):

We compare forecasts using **errors** (how wrong was the guess?).
Common easy measure: **MAE (Mean Absolute Error)** = average of |Actual – Forecast|

Lower MAE = better method for that data.

1. What happens if we use 2-month or 4-month Moving Average instead of 3-month?

(Impact on forecast + error)

Simple rule:

- **Shorter window** (2-month MA) = **reacts faster** but can be **jumpy**.
- **Longer window** (4-month MA) = **smoother & calmer** but **lags behind** more.

Real numbers (calculated on the same 12-month ice-cream data):

Method	**Mean Absolute Error (MAE)**	**How it behaves on this data**

2-Month MA	**28.75**	**Best** – catches summer spike quickly
3-Month MA	33.33	**Medium** – balanced but slower
4-Month MA	**38.75**	**Worst** – lags a lot during big jumps

Why this happens (real-life picture):

Your ice-cream sales suddenly jump from 130 (Apr) → 160 (May) → 200 (Jun) because of summer.

- **2-month MA** looks only at the last 2 months → it quickly sees the rising trend and forecasts higher numbers sooner → lower error.
- **The 4-month MA** looks at the last 4 months (which still include low winter numbers) → it keeps giving lower forecasts even when summer has already started → bigger mistakes (higher error).

2. How is the alpha (α) value decided?

Hit-and-trial or fixed based on history?

Short answer:
Mostly by hit-and-trial (trial & error) on past data — not fixed forever.

Alpha is **customized per product/SKU** based on its history.

How it actually works (simple steps):

1. Take the last 12–24 months of **historical demand**.
2. Try different alpha values: 0.1, 0.2, 0.3, …, 0.8, 0.9.
3. For each alpha, run Exponential Smoothing and calculate MAE (or other error).
4. Pick the alpha that gives the **lowest error**.

Real-life example:

- A new trendy phone (demand jumps a lot) → best alpha might be **0.7 or 0.8** (reacts very fast).
- A regular milk packet (demand very stable) → best alpha might be **0.1 or 0.2** (keeps it calm).

In big companies they don't do it manually every time — software (like SAP, Oracle, or even Python) automatically tests 0.01 to 0.99 and picks the best one for each SKU every month.